

UNIT-4 | Lecture 5

Dependency Preservation & Lossless Design

Coverage: Hour 5

Learning Objectives

Understand dependency preservation and lossless-join decomposition.

Differentiate the two design properties.

Dependency Preservation

A decomposition is **dependency-preserving** if the functional dependencies of the original relation can be enforced by checking the decomposed relations individually.

This makes constraint checking simpler and avoids unnecessary joins.

Lossless Design

A decomposition is **lossless** (non-additive) if the natural join of the decomposed relations produces exactly the original relation.

A **lossy** decomposition can generate spurious tuples that were not in the original relation.

Simple Idea

Original relation: $R(A, B, C)$. Decompose into $R_1(A, B)$ and $R_2(A, C)$. Whether this decomposition is lossless depends on the functional dependencies and the common attributes; a correct decomposition must be verified rather than assumed.

Comparison

Property	Meaning
Lossless join	Join reconstructs the original relation exactly; no spurious tuples.
Dependency preservation	Original FDs can be enforced from the decomposed relations.

Design Principle

During normalization, aim for a **lossless** decomposition and, where possible, a **dependency-preserving** decomposition.

Standard 3NF synthesis can provide lossless and dependency-preserving decompositions; BCNF decomposition can sometimes lose dependency preservation.

Key Takeaway

Lossless design protects information during decomposition, while dependency preservation keeps functional dependencies enforceable.